Alex Tyh Wang

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Summary

Computer science undergraduate with experience architecting **customer-facing AI-powered solutions**, **agent tooling**, and **production software systems**. Passionate about **end-to-end ownership** and delivering cross-functional, highly scalable technical solutions to complex and ambiguous engineering challenges.

Education

Rochester Institute of Technology — Rochester, NY
Bachelor of Science in Computer Science — Expected May 2028

Skills

Languages: Python, Java, C++, TypeScript, JavaScript, C, C#, SQL
Frameworks and Tools: React, Angular, Next.js, Spring Boot, REST, Node.js, PostgreSQL, Azure DevOps, Jenkins
AI and Systems: Agentic engineering, prompt engineering, context engineering, LLM evaluation, AI agents, multi-agent systems, subagent architectures, MCP, tool calling, RAG, model reliability, OpenAI API, OpenRouter

Experience

Software Engineer Intern — Jan 2026 – Aug 2026
Paychex — *Rochester, NY*

- Led end-to-end development of an internal developer platform on the **AI Integrations** team, directly collaborating with stakeholders to integrate **AI agents and MCP tooling** into enterprise workflows; presented the platform to executives.
- Built LLM evaluation systems for real developer workflows, comparing model behavior, reliability, and usefulness.
- Authored a **subagent architecture** to reduce token usage by **55%** and optimize workflow performance for high-stakes internal tools.

Software Engineer Intern — Feb 2025 – Aug 2025
Real Wave Capital — *Remote*

- Architected production-grade Angular/TypeScript UIs for autonomous **AI agents** and rewrote the MCP tool-calling and context strategy, managing complete end-to-end execution to support 100+ daily users and reduce token spend by roughly **50%**.
- Integrated OpenAI API and OpenRouter into production agent flows with .NET backend contributions, improving endpoint response times by **25%** and scaling throughput to **6,000+ weekly requests**.

Projects

Chatio Agentic Behavior Benchmark and Chat App — Jun 2025 – Present
Next.js, Python, TypeScript — *chatio.dev*

- Built an end-to-end benchmark platform for LLMs measuring real-world usability through task-driven tests.
- Created a production-grade streaming chat interface using Next.js 15 with real-time SSE streaming, multi-step agentic workflows, tool calling, and dynamic parameter optimization.

iMessages Chatbot Server — Nov 2025 – Present
Python, asyncio, SQLite, AppleScript — *GitHub*

- Built an asyncio Python API gateway for macOS Messages, supporting up to 20 concurrent stateful AI conversations over iMessage.
- Engineered iMessage AI infrastructure with SQLite conversation memory, 0.3s debounce batching, and semaphore-controlled LLM API calls.

Word Hunt — Sep 2024 – Dec 2024
Next.js, TypeScript, Figma — *wordhunt.org*

- Built and launched a Next.js word game that reached 250K+ impressions and 3.9K organic clicks.

Leadership and Activities

Founder and President — Jan 2022 – Present
Bay Area Stem Academy — *San Jose, CA | bayareastemacademy.org*

- Founded a 501(c)(3) nonprofit, communicating cross-functionally to teach **660+ students** through **15+ free camps** and partner with **5+ schools**.
- Owned complete execution of website development, curriculum, mentoring, outreach, and logistics, providing **$35,000+** in free STEM education and donating **$5,000+** to partner organizations and teams.
- Initiated a collaboration with **SanDisk** and co-hosted a volunteer day at SanDisk's Milpitas campus, mobilizing employees to build STEM kits for future programs.